

Total No. of Questions : 8]

SEAT No. :

PE-4266

[Total No. of Pages : 2

[6582]-38

S.E. (Computer Engineering)

MICROPROCESSOR

(2019 Pattern) (Semester - IV)(210254)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Black Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.

Q1) a) Explain following in detail [6]

- i) Global Descriptor Table
- ii) Interrupt Descriptor Table

- b) Draw and explain segment selector format. [6]
- c) With the necessary flowchart, explain the complete address translation process in 80386. [6]

OR

Q2) a) Explain the use of following instructions in detail. [6]

- i) SGDT
- ii) LLDT
- iii) SIDT

- b) Draw and explain the general descriptor format. [6]
- c) With the necessary diagram, explain the logical to linear address translation process of 80386. [6]

Q3) a) Explain different levels of protection? State the rules of protection check. [6]

- b) Write a short note on CPL, DPL, and EPL. [6]
- c) Explain conforming and non-conforming code segments. [5]

OR

P.T.O.

- Q4)** a) Explore the need for a protection mechanism in 80386. [6]
b) Define the functions of Type Checking and Limit Checking in protection. [6]
c) Explore the role of various fields in page level protection. [5]

- Q5)** a) Define task switching and explain the steps involved in task switching operation. [6]
b) Explain the TSS descriptor of 80386 with a neat diagram. [6]
c) Differentiate between real mode and virtual mode. [6]

OR

- Q6)** a) Draw and explain task gate descriptor of 80386. [6]
b) Draw and Explain the Task State Segment of 80386. [6]
c) With the necessary diagram, explain entering and leaving the virtual mode of 80386. [6]

- Q7)** a) With the help of neat diagram explain the process of handling interrupts in protected mode. [6]
b) Differentiate and Explain the Interrupt gate and Trap gate descriptor. [6]
c) Explain various features of the 8051 microcontroller. [5]

OR

- Q8)** a) Explain the following exception conditions with an example. Faults, Traps, and Aborts. [6]
b) What is the error code? Explain with the necessary diagram. [6]
c) List and elaborate on different applications of microcontrollers. [5]

